General-purpose depth on C3VD: evaluation and target-domain calibration

- One frozen released model, not a new network
- Development sequence fits the baseline and two calibrations
- Test sequence is scored once, after the parameters are frozen

Distance from a single endoscopic image is ambiguous

- Camera-to-surface distance supports spatial awareness in medical robotics
- C3VD pairs colonoscope video of a physical model with registered depth
- RGB is a phantom recording; ground truth is rendered, not a patient scan
- The question is transfer of a general model, not clinical validation

Development fits parameters. Test only evaluates them.

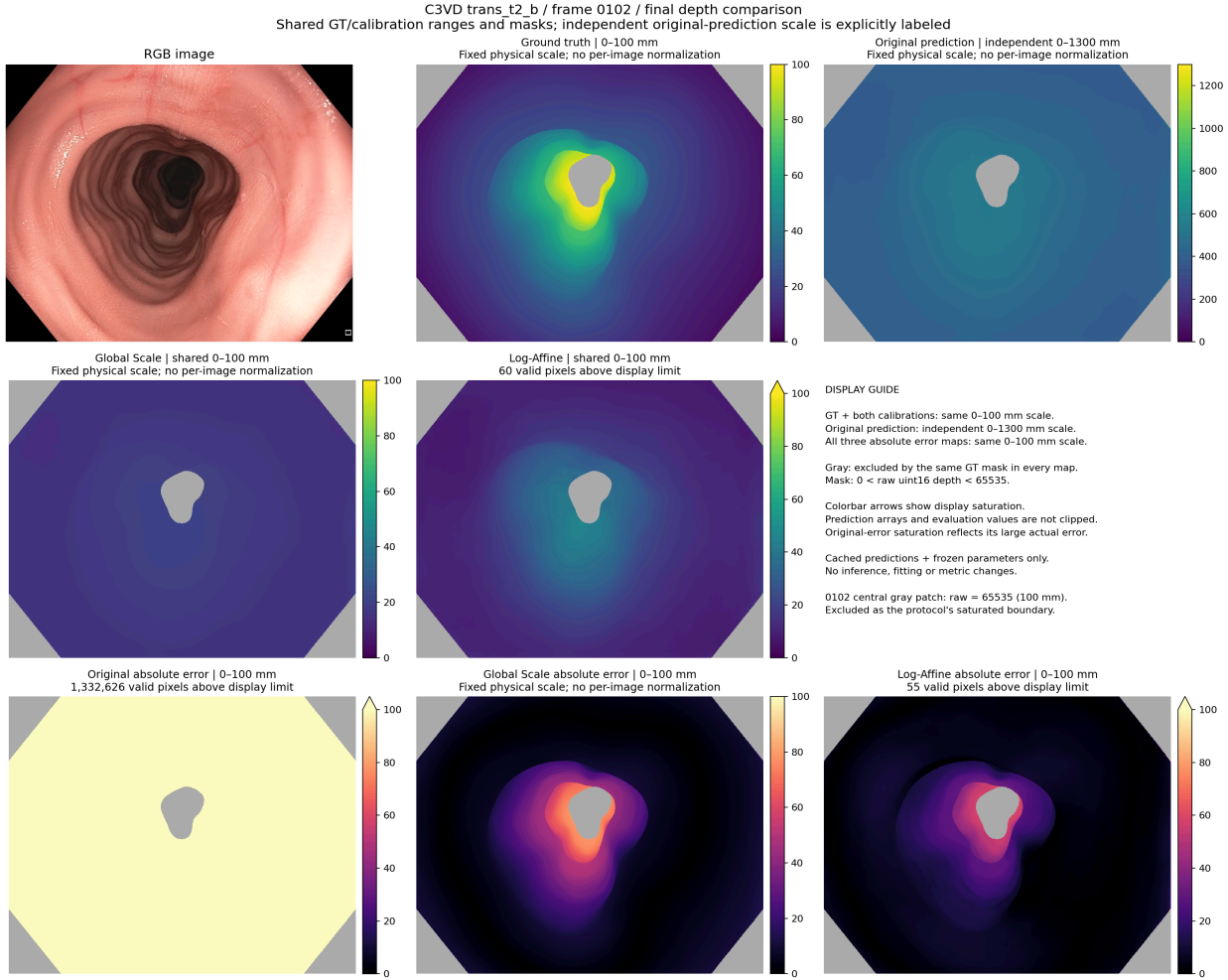
- `trans_t1_a`: 61 frames `trans_t2_b`: 103 frames
- Depth in mm: raw × 100 / 65535, valid where 0 < raw < 65535
- 136,655,314 valid test pixels, pooled, not averaged as frames
- AbsRel, RMSE (mm), and the fraction with ratio strictly below 1.25

Same mask, same pixels, different use of the image

- Constant median: 17.2595 mm everywhere. No image structure
- Original Depth Anything V2: frozen indoor metric checkpoint
- Global scale: multiply by $s = 0.0382144$, fit on development only
- Log-affine: $\exp(a + b \ln D)$, $a = -26.3679$, $b = 4.7915$

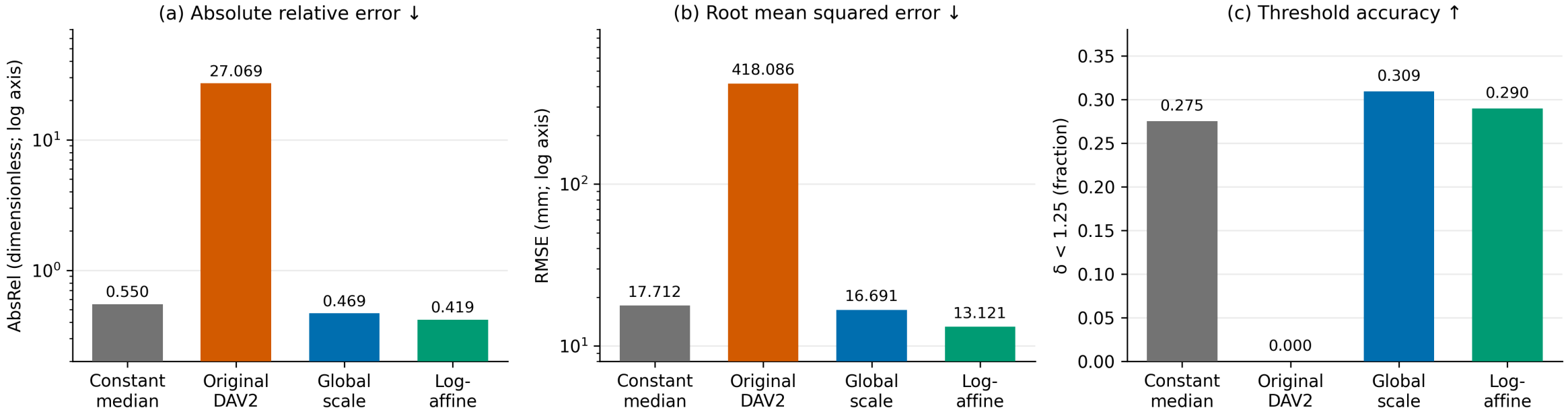
Only the original checkpoint is zero-shot relative to our development labels.

Test frame 0102, chosen as the last frame



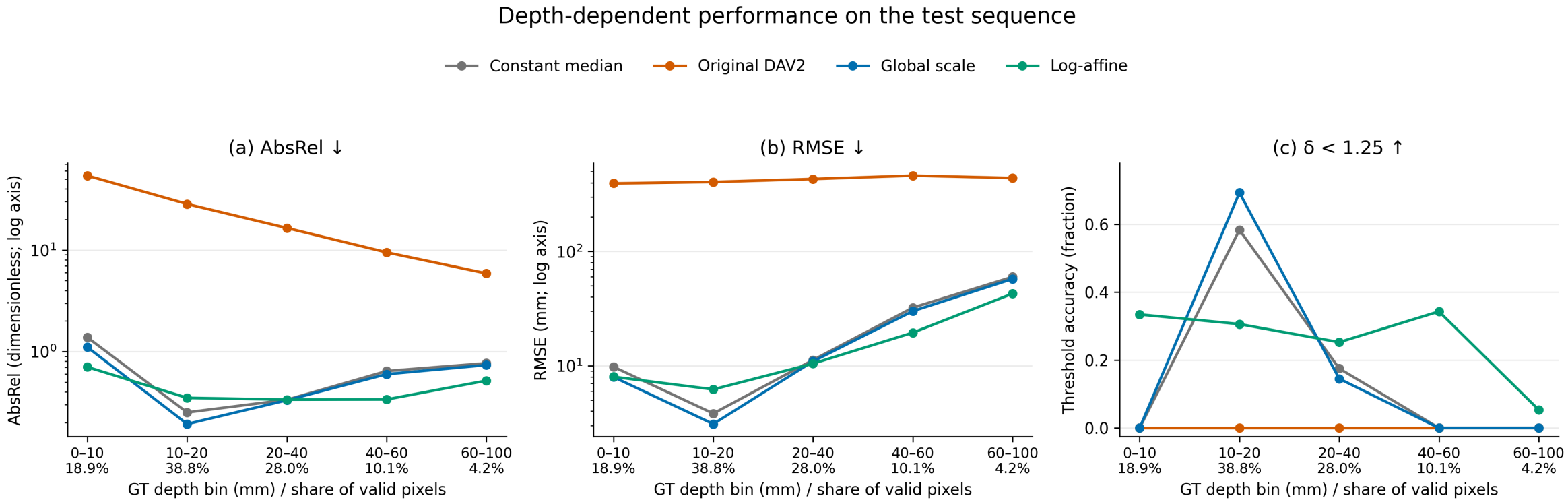
Log-affine wins AbsRel and RMSE. Global scale wins δ.

Four-method comparison on the held-out test sequence



All 136,655,314 valid test pixels pooled; exact scores are in Table 1. Zero δ is shown at the baseline.

The δ loss is concentrated at 10–20 mm



Bins are left-closed and right-open. All valid pixels are retained; scores are pooled within each bin.

A lower average error is not more accurate pixels

Method	AbsRel	RMSE	δ < 1.25
Constant median	0.5499	17.71 mm	0.2752
Original DAV2	27.0687	418.09 mm	0
Global scale	0.4685	16.69 mm	0.3092
Log-affine	0.4191	13.12 mm	0.2897

Relative to global scale: AbsRel −10.54%, RMSE −21.39%, δ −1.96 percentage points.

What these numbers do not say

- Not label-free: baseline and both calibrations use development depth
- Not 103 patients: one continuous test video
- Not recovered folds: Pearson 0.784 is large-scale association
- 60–100 mm remains too shallow: mean GT 76 mm, log-affine about 36 mm

Calibration changed the scale. It did not finish the geometry.

- The released model had relative structure and a severe scale error
- One development scale factor beat a constant depth
- Log-affine improved magnitude error and paid a near-range δ penalty
- Report that trade-off. Do not report a new algorithm or a clinical system